These equipment should be designed, tested, instal led and maintained based on these standards, taking into account that th e purpose of Navigation Lights is to identify ships and to notify their

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SOURCE: MSC.253(83).pdf
These equipment should be designed, tested, instal led and maintained based on these standards, taking into account that th e purpose of Navigation Lights is to identify ships and to notify their intentions at sea and that the purpose of a Navigation Lights Controller is to provide means of control and monitoring of the stat us of navigation lights onboard th e vessel to the Officer of the Watch (OOW).

2 Application

In addition to the general requirements set out in resolution A.694(17) 1, navigation lights, navigation lights controllers and associated eq uipment should meet the requirements of these standards.

3 Definitions

3.1 Associated equipment means equipment necessary for the operation of NLs and NLCs.

3.2 COLREGs means Convention on the In ternational Regulations for Preventing Collisions at Sea, 1972, including their annexes.

3.3 Lamp means a source producing light, includi ng incandescent sources, Light Emitting Diodes (LED) and other non-incandescent sources.

3.4 Length means the length overall.

3.5 Navigation Light (NL) means the following lights:

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SOURCE: MSC.253(83).pdf
3.3 Lamp means a source producing light, includi ng incandescent sources, Light Emitting Diodes (LED) and other non-incandescent sources.

3.4 Length means the length overall.

3.5 Navigation Light (NL) means the following lights:

.1 masthead light, sidelights, sternlight, towing light, all-round light, flashing light as defined in Rule 21 of COLREGs;

.2 all-round flashing yellow light required for air-cushion vessels by Rule 23 of COLREGs; and

.3 manoeuvring light required by Rule 34(b) of COLREGs.

The light source includes lamps, its housing, placing and means for delimiting the angle of lighting.

1 Refer to publication IEC 60945.

3.6 Navigation Light Controller (NLC) means a device enabling operational control of a Navigation Light.

3.7 SOLAS means the International Convention for the Safety of Life at Sea, 1974, as amended.

4 Navigation Lights

4.1 General

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SOURCE: MSC.253(83).pdf
3.6 Navigation Light Controller (NLC) means a device enabling operational control of a Navigation Light.

3.7 SOLAS means the International Convention for the Safety of Life at Sea, 1974, as amended.

4 Navigation Lights

4.1 General

4.1.1 Unless expressly required otherwise, NLs should appear steady and non-flashing.

4.1.2 Lenses of NLs should be produced in a robust, non-corroding material, which should ensure a long-term durability for the optical qualities of the lens.

4.1.3 A masthead light, sidelights a nd a sternlight installed on bo ard a ship not less than 50 m in length should be duplicated or be fitted with duplicate lamps.

4.1.4 Only lamps specified by the manufacturer shoul d be used in each particular NL to avoid reduction of NL’s performance due to unsuitable lamps.

4.1.5 A sufficient number of spare lamps for NLs should be carried onboard, taking into account the duplication of NLs or lamps, as appropriate.

4.2 Luminous intensity distribution

4.2.1 In the horizontal directions where decrease of luminous intensity to “practical cut-off” is required by section 9 of Annex I to COLREGs, the luminous intensity should be no more than

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SOURCE: MSC.253(83).pdf
4.2 Luminous intensity distribution

4.2.1 In the horizontal directions where decrease of luminous intensity to “practical cut-off” is required by section 9 of Annex I to COLREGs, the luminous intensity should be no more than 10% of the average luminous intensity within the prescribed sector for vessels not less than 12 m in length.

4.2.2 Within the prescribed sector in which th e minimum luminous intensity is required by section 9 of Annex I to COLREG s, the horizontal inte nsity distribution of the light should be uniform in such a way that the measured mi nimum and maximum luminous intensity values (in candelas) do not differ by more than a fact or of 1.5, to avoid luminous intensity changes which may result in the appearance of a flashing light for vessels not less than 12 m in length.

4.2.3 Within the prescribed sector in which th e minimum luminous intensity is required by section 10 of Annex I to COLREG s, the vertical intensity distribution of the light should be uniform in such a way that the measured mini mum and maximum luminous intensity values (in candelas) do not differ by more than a factor of 1.5, to avoid luminous intensity changes which

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SOURCE: MSC.253(83).pdf
uniform in such a way that the measured mini mum and maximum luminous intensity values (in candelas) do not differ by more than a factor of 1.5, to avoid luminous intensity changes which may result in the appearance of a flashing light for vessels not less than 12 m in length.

4.3 Special requirements for lights using LEDs

The luminous intensity of LEDs gradually decreas es while the electricity consumption remains unchanged. The rate of decrease of luminous intensity depends on the output of LEDs and temperatures of LEDs. To prevent shortage of luminous intensity of LEDs:

.1 An alarm function should be activated to notify the Officer of the Watch that the luminous intensity of the light reduces below the level required by COLREGs;

or

.2 LEDs should only be used with in the lifespan (practical term of validity) specified by the manufacturer to maintain the nece ssary luminous intensity of LEDs. The

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SOURCE: MSC.253(83).pdf
or

.2 LEDs should only be used with in the lifespan (practical term of validity) specified

by the manufacturer to maintain the nece ssary luminous intensity of LEDs. The lifespan of LEDs should be determined and clearly notified by the manufacturer based on the appropriate test results on the decrease of luminous intensity of the LEDs under various temperature conditions and on the temperature condition of LEDs in the light during operation, taking the appropriate margin into account.

5 Navigation Light Controller

5.1 An NLC should facilitate ON/O FF controls of individual NLs.

5.2 An NLC should provide vi sual indications of “ON”/“OFF” status of NLs.

5.3 Pre-programmed NL group settings may be provided.

5.4 An NLC on board a ship not less than 50 m in length should provide the alarm for:

.1 failure of power supply to NLs; and

.2 failure, including short circuit, of a lamp which is switched ON.

5.5 An NLC on board a ship not less than 50 m in length should present the status of all NLs in a logical presentation, meeting the requ irements set out in resolution MSC.191(79), e.g ., by symbol marks on a display.